

[Courseware \(/courses/HumanitiesScience/StatLearning/Winter2014/courseware\)](/courses/HumanitiesScience/StatLearning/Winter2014/courseware)[Course Info \(/courses/HumanitiesScience/StatLearning/Winter2014/info\)](/courses/HumanitiesScience/StatLearning/Winter2014/info)[Discussion \(/courses/HumanitiesScience/StatLearning/Winter2014/discussion/forum\)](/courses/HumanitiesScience/StatLearning/Winter2014/discussion/forum)[Wiki \(/courses/HumanitiesScience/StatLearning/Winter2014/course_wiki\)](/courses/HumanitiesScience/StatLearning/Winter2014/course_wiki)[Progress \(/courses/HumanitiesScience/StatLearning/Winter2014/progress\)](/courses/HumanitiesScience/StatLearning/Winter2014/progress)[Lecture Slides \(pdf\) \(/courses/HumanitiesScience/StatLearning/Winter2014/ba1951b8f66c4cdca2fcaabdc91b792/\)](/courses/HumanitiesScience/StatLearning/Winter2014/ba1951b8f66c4cdca2fcaabdc91b792/)[R Sessions \(/courses/HumanitiesScience/StatLearning/Winter2014/0d68641c19484ef9aa6aeea03426dc68/\)](/courses/HumanitiesScience/StatLearning/Winter2014/0d68641c19484ef9aa6aeea03426dc68/)

9.1.R1 (1/1 point)

If β is not a unit vector but instead has length 2, then $\sum_{j=1}^p \beta_j X_j$ is

- ☒ twice the signed Euclidean distance from the separating hyperplane $\sum_{j=1}^p \beta_j X_j = 0$ ✓
- ☐ half the signed Euclidean distance from X to the separating hyperplane
- ☐ exactly the signed Euclidean distance from the separating hyperplane

Explanation

We know $\beta' = \frac{1}{2} \beta$ has length 1, so it is a unit vector in the same direction as β . Therefore, $\sum_{j=1}^p \beta_j X_j = 2 \sum_{j=1}^p \beta'_j X_j$, where $\sum_{j=1}^p \beta'_j X_j$ is the Euclidean distance.

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